**Code No: 158AK** **R18**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year II Semester Examinations, July/August - 2022**
**COMPUTATIONAL COMPLEXITY**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max.Marks:75**

**Answer any five questions**
**All questions carry equal marks**
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1.a) Explain Polynomial time and its justification.
b) What is the P versus NP problem and why it's hard? [8+7]

2. What is Polynomial Time? Explain with Nontrivial examples of polynomial-time Algorithms. [15]

3.a) Illustrate Matrix Chain Multiplication with an example.
b) Explain how knapsack problem is solved using Dynamic Programming. [8+7]

4.a) Illustrate the implementation of Quick Sort.
b) Explain how travelling sales person problem is solved using Branch-and-bound technique. [8+7]

5. What are Randomized Algorithms? Explain in detail about Finger Printing and Pattern Matching. [15]

6. Write a detailed note on Graph problems. [15]

7. Explain in detail about Approximation Algorithms and Polynomial Time Approximation Schemes. [15]

8.a) Explain in detail about Decision Trees and Circuits.
b) Write a short note on Disjoint Sets. [8+7]

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